

Plastic waste analysis - Blantyre, Malawi

Testing Document

Version 1

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General description and purpose

Purpose of test

To evaluate which separation steps should be implemented, the existing mixed plastic waste stored in Blantyre, Malawi must be analyzed.

Important criteria are: Present plastic types and flakes sizes.

The mixed plastic is stored in 1200 bags (estimate) of roughly 30 kg of plastic each.

Description of Test

First, samples are taken from the stored plastic. Since the waste in the bags is not homogeneous, the content of multiple bags is sampled. The bag is cut open and a small sample of flakes is gathered (choose different looking/colored flakes if possible). Then, the opening is sealed with duct tape and the bag as well as the sample are marked with an ID number to allow tracking. This process is repeated multiple times.

Once the samples are collected, the flakes are washed in a water-dish soap mixture and dried. Lastly, single flakes from each sample are scanned with the handheld NIR spectrometer ([MicroPHAZIR PC](#)). Hereby, flakes exhibiting differences in color, size, or floating/sinking behavior in water are chosen and their plastic type is recorded. Limitations of the NIR spectrometer are the missing distinguishability between LDPE and HDPE (MicroPhazir only returns PE) and the inability to classify dark flakes.

Calculations

None.

Schedule of test with all steps required

1. Cut open a bag of mixed plastic waste.
2. Collect a small sample of flakes. Choose flakes of different color or size if applicable.
3. Close the opening with duct tape.
4. Mark the bag and the sample with its ID.
5. Repeat steps **1-4** for at least **5 bags** at different locations in the pile. (Only bags that can be reached easily!)
6. Create a bath of water and dish soap.
7. Wash the flakes by hand one sample at a time.
8. Let the flakes dry.
9. For each sample, pick out flakes exhibiting differences in color, size, and/or floating/sinking behavior in water and scan the material with the NIR spectrometer.

Risk analysis

Potential risks associated with this test

The origin and content of the bags is unknown. Harmful dust particles could be present and dispersed in the air while sampling.

The bags have become very fragile over the long storing period and rip easily, possibly resulting in an avalanche like material stream.

Sharp objects could hide within the material. Small particles could get into the eyes.

Mitigation steps

A face mask (FFP3 recommended, minimum FFP2), safety glasses, and gloves are worn during the sampling. Furthermore, high boots are worn to improve stability and protect the ankles from falling material.

Only bags that are directly accessible without shifting around bags or climbing the pile are sampled.

Risks after taking mitigation steps

Ripping of bags and loosening of large amounts of plastic flakes is still possible. The sampling must be handled with care.

Tools and devices required

Tool	Check-in tick
1x safety glasses	X
1x pair of gloves	X
1x face mask (min. FFP2, rec. FFP3)	X
1x pair of boots	X
1x key to storage	X
1x knife	X
Plastic bags (samples)	X
Duct tape	X
1x marker	X
Water to wash hands	X
Dish soap	X
Water	X
1x container for cleaning	X
1x MicroPHAZIR PC	X
1x laptop	X
1x camera (phone)	X

Testing Results

All pictures of the process and the results for each sample are available [here](#).

Only already open bags were sampled to limit the amount of plastic flakes spreading in the facility, since the pollution of the bags precluded resealing with duct tape.

Six bags were sampled. The extracted and washed flakes are shown in Figure 1.



Figure 1: Plastic flake samples from the storage in Blantyre, Malawi.

ID	Observations	Material types (scanned)	Flake sizes [mm]
1	(almost) fully floating; little debris on bottom; water almost clear; fibers of bag in sample	all PE (likely HDPE?)	1x1 up to 23x4 & 19x9
2	(almost) fully sinking; water little turbid; fibers of bag in sample	mostly PET , one PP (fibers of bag?), one Unidentified (black)	1x1 up to 15x8, many <1 mm
3	almost fully sinking; many very small pieces; water little turbid	all PET (including one foil piece)	1x1 up to 29x6
4	mostly sinking, some floating; water turbid; fibers (grass/bag?) in sample; twig in sample	float: mostly PP , two flakes PE (likely HDPE?) sink: mostly PET , one PE (likely HDPE?), one Unidentified (brown, maybe PET or PETG)	1x2 up to 33x7
5	fully sinking; relatively clear water	mostly PET , one PE (foil = LDPE?)	2x2 up to 20x9
6	(almost) fully sinking; small plastics floating (foils?); relatively clear water; small particles in water	mostly PET , one PP , one Unidentified (foil)	1x2 up to 18x19, some smaller than holes in strainer

The scanned flakes from sample 4 are shown in Figure 2.

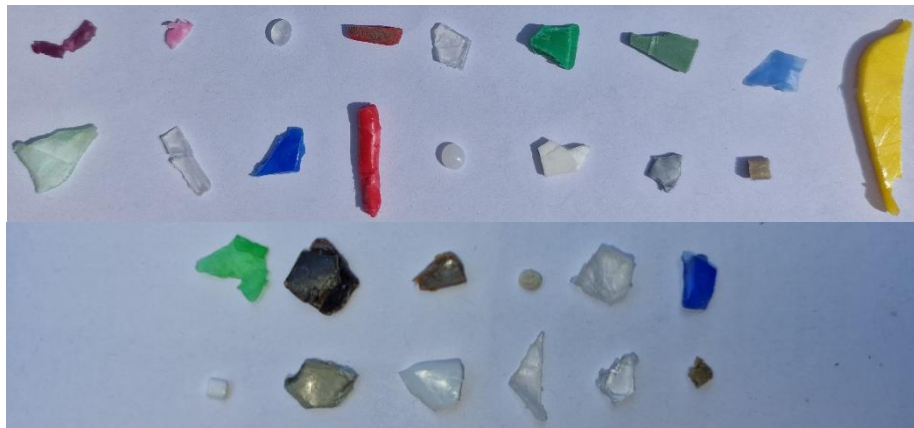


Figure 2: The scanned flakes from sample 4. Top: floating in water; bottom: sinking in water.

Evaluation

The plastic bags appear to already have undergone some sorting process. Except for sample 4, all bags show one predominant plastic type.

The identified plastic types present in the samples are: LDPE and/or HDPE, PP, PET. Unidentifiable flakes can be attributed to the dark color of the flake (black or brown) inhibiting the classification by the NIR spectrometer or an unknown or impure plastic type.

A wide range of flake sizes has been identified reaching from particles smaller than the strainer hole size up to 350 mm².

Due to the partially large pollution of the bags and the flakes, a washing step must precede the sorting.

Since bags located further back or towards the bottom of the pile could not be reached, the presence of other plastic types cannot be ruled out.

Bibliography